

ISE549 Week 1 Management of Innovation and Technology

Lecture: Knowledge and Innovation in the age of digitalisation, basic concepts in Strategies and Business Management

→ How firms create value through innovation, adapt to technological change, manage data and disruption, and avoid becoming trapped by past success.

1. Course Big Picture: Innovation as a Continuous System

Core idea

English	中文
A modern firm does not operate in a stable environment. It must continually respond to markets, technology, competition, and new opportunities.	現代企業並不是在穩定環境中運作；它必須持續回應市場、科技、競爭與新機會。
The course views innovation through a continuous loop: ideas → value creation → market competition → new challenges/opportunities → new ideas.	課程把創新視為一個持續循環：想法 → 創造價值 → 市場競爭 → 新挑戰／機會 → 新想法。
Technology, management, and innovation work together rather than separately.	科技、管理與創新並不是分開運作，而是互相配合。

The lecturer's key message is that firms need to keep the whole system in view, rather than focus only on one product, one technology, or one short-term decision. This is the idea of "seeing the woods for the trees."

Lecture argument

- Firms survive by maintaining the loop, not by treating innovation as a one-off project.
企業能否生存，取決於能否維持這個循環，而不是把創新當作一次性的專案。
- An idea becomes meaningful only when it is transformed into value that customers, society, or the market can accept.
一個想法只有轉化成客戶、社會或市場願意接受的價值時，才真正有意義。
- Innovation later needs a business model, market launch, and strategic response to competition.
創新之後還需要商業模式、市場推出策略，以及面對競爭的回應。

Ideas must become value; value must survive the market.

Exam takeaway

If asked to explain innovation in a managerial context, do not define it as merely "having a new idea." Emphasize the process of turning ideas into accepted value and sustaining that value under market competition.

若題目問「管理脈絡下的創新」，不要只寫「有新想法」；應強調把想法轉化成被市場接受的價值，並在競爭中持續維持這種價值。

2. Systems Thinking and the Importance of Context 系統思維與情境的重要性

Definition

English	中文
Systems thinking means understanding a situation as a set of interconnected elements rather than isolated parts.	系統思維是把情況理解為一組相互連結的元素，而不是互不相關的部分。
A system involves inputs, processes/systems, outputs, and feedback loops.	一個系統包括輸入、過程／系統、輸出與回饋循環。
A change in one part can influence the whole system.	一個部分的改變可以影響整個系統。

The materials stress that **systems are made of interactions and feedback loops**. Managers therefore need to evaluate relationships and consequences, not simply make isolated decisions.

- No business theory is universally true in every context.
沒有任何商業理論能在所有情境下永遠成立。
- “Everything affects everything”: when assessing a concept, strategy, or innovation, managers should revisit the external and internal driving forces that may affect it.
「所有事情都會互相影響」：在評估一個概念、策略或創新時，管理者應重新檢視可能影響它的內外部驅動力。

Example

A disruption in a shipping port can affect raw-material deliveries, factory schedules, energy use, customer delivery promises, and pricing decisions at the same time.

一個港口受干擾，可同時影響原材料交付、工廠排程、能源使用、向顧客承諾的送貨時間，以及定價決策。

Exam takeaway

When analyzing a case, show how different elements connect. For example:

- External shock → supply chain disruption
外部衝擊 → 供應鏈中斷
- Supply chain disruption → production adjustment
供應鏈中斷 → 生產調整
- Production adjustment → customer delivery and pricing impact
生產調整 → 顧客交付與定價受影響

3. Global Trends Driving Innovation 推動創新的全球趨勢

3.1 Accelerating technology adoption 科技採用速度加快

English	中文
Technology adoption has become much faster than in earlier generations.	科技採用速度比早期世代快得多。
The telephone took about 60 years to reach widespread household adoption, while the internet reached 50 million users in about 7 years.	電話大約用了 60 年才廣泛進入家庭；互聯網大約 7 年便達到 5,000 萬使用者。
Faster adoption reduces traditional market barriers and forces firms to adapt more quickly.	更快的採用速度會降低傳統市場障礙，迫使企業更迅速適應。

The lecturer frames modern adoption as a “viral sprint” rather than a slow, generational process.

3.2 The Fourth Industrial Revolution 第四次工業革命

Industrial Revolution	Main feature	中文重點
First Industrial Revolution	Steam and water-powered mechanisation	蒸汽與水力推動機械化
Second Industrial Revolution	Electricity and mass production	電力與大量生產
Third Industrial Revolution	Computers and automation	電腦與自動化
Fourth Industrial Revolution	Cyber-physical systems, IoT, AI, real-time integration	資訊物理系統、物聯網、AI、即時整合

A **cyber-physical system** tightly integrates physical operations with digital networks so that the system can monitor, communicate, and adjust in real time.

資訊物理系統（**cyber-physical system**）把實體營運與數碼網絡緊密整合，使系統可即時監察、溝通與調整。

Example : Smart factory / 智慧工廠

If a typhoon disrupts a major shipping port: 若颱風令主要港口運作受阻

1. Weather data enters supply-chain software. 天氣數據進入供應鏈軟件
1. The system predicts delayed materials. 系統預測原材料將延遲
2. Factory machines automatically adjust production and conserve resources.
工廠機器自動調整生產及節省資源
3. Customer-facing systems update delivery estimates and possibly prices.
面向顧客的系統更新送貨時間，甚至調整價格。

This illustrates real-time self-optimisation with limited human intervention.

這展示了在有限人手干預下的即時自我優化。

3.3 Data abundance and AI 數據豐裕與人工智能

English	中文
The modern managerial problem is no longer only information scarcity; it is information abundance.	現代管理難題不再只是資訊不足，而是資訊過量。
Data was described as the “fuel” of cyber-physical integration and AI-driven decision-making.	數據被形容為資訊物理整合與 AI 驅動決策的「燃料」。
AI becomes important because it can identify patterns and extract intelligence from large, complex data sets.	AI 變得重要，是因為它能從大量複雜數據中找出模式並提取洞見。

Data is not value by itself; intelligence creates value.

數據本身不等於價值；洞見才會創造價值。

Exam takeaway

For a digital transformation question, connect the logic:

Data → Analytics / AI → Better decisions → Operational and customer value

4. Customer Satisfaction vs Customer Experience 顧客滿意度與顧客體驗

Definition

Concept	English	中文
Customer satisfaction	A transactional judgment: did the product or service perform its expected function?	一種交易式判斷：產品或服務是否完成預期功能？
Customer experience	The end-to-end, emotional and relational journey a customer has with a company.	顧客與企業互動時所經歷的端到端、情感性及關係性旅程。

Customer satisfaction alone is vulnerable because customers can easily switch if a competitor offers a similar product at a lower price. Customer experience is more defensible because it creates loyalty through the entire journey.

單靠顧客滿意度其實很脆弱，因為競爭者只要提供類似但更便宜的產品，顧客便可能轉走。顧客體驗較具防禦力，因為它透過整個旅程建立忠誠度。

Example / 例子：Hammer store / 賣鎚子的商店

- A hammer that works properly may create satisfaction.
一把能正常釘釘子的鎚子可以帶來滿意度。
- But if another store sells an identical hammer for less, the customer may switch immediately.
但如果另一間店以更低價售賣相同鎚子，顧客可能立即轉向。
- Experience includes app usability, delivery smoothness, customer-service tone, and emotional connection to the brand.
體驗則包括 app 是否易用、送貨是否順暢、客服語氣，以及與品牌的情感連結。

Exam takeaway

Satisfaction = functional performance.

Experience = total journey + emotion + relationship

In a case answer, explain how customer experience can become a competitive advantage by creating loyalty and lowering customers' willingness to switch.

在案例題中，可說明顧客體驗如何透過建立忠誠度、降低顧客轉換意願，形成競爭優勢。

5. Value Creation and the Three Value Disciplines 價值創造與三種價值紀律

Definition

The lecture identifies three competing value disciplines that firms may pursue to create value.

Value discipline	English explanation	中文說明
Operational Excellence	Deliver products/services efficiently, reliably, and at an attractive cost.	以高效率、可靠性及具吸引力的成本提供產品／服務。
Product/Service Leadership	Offer leading-edge, superior, or highly innovative products/services.	提供領先市場、優質或高度創新的產品／服務。
Customer Intimacy	Build deep customer knowledge and tailor offerings closely to customer needs.	深入理解顧客，並高度配合其需要來設計產品／服務。

Operational excellence was linked with the “cheap-good-fit for purpose” idea: customers receive something affordable, good enough in quality, and suitable for its intended use.

營運卓越與「平、靚、正／便宜、好用、合用途」的概念相關：顧客得到價格合理、質素足夠且符合用途的產品。

Exam takeaway

When discussing strategy, identify which discipline the company prioritizes and explain the trade-off. A firm usually needs a clear primary discipline rather than trying to be best at everything.

討論策略時，要指出企業優先追求哪一種價值紀律，並解釋取捨。企業通常需要有清晰的主導方向，而不是每一方面都想做到最好。

6. Innovation Process: From Ideas to Market Value 創新過程：從想法到市場價值

Core process

Ideas → Insight / Problem Identification → Solutions → Refinement / Iteration
→ Business Model → Market Launch

The lecturer describes innovation as a step-by-step process: ideas are developed, solutions are refined, and the offering must eventually be taken to market through a business model. 講者指出，創新是一個循序漸進的過程：從發展想法、持續優化解決方案，到最終透過商業模式推向市場。

Important distinction

English	中文
Innovation is incremental as a process: it often develops step by step through iteration.	創新作為一個過程具有漸進性：它通常經過一步一步的迭代發展。
Incremental innovation is a specific type of innovation involving relatively small changes.	漸進式創新則是一種特定創新類型，指相對小幅度的改變。

Do not confuse these two uses of “incremental.”

不要混淆「創新過程是逐步推進」與「漸進式創新」這兩種意思。

Tools along the process

Stage	English	中文
Insight and idea generation	Creativity and ideation	創意與構思
Problem and solution design	Open innovation; iterative development	開放式創新；迭代開發
Testing and refinement	Prototyping, feedback, iteration	原型、回饋、迭代
Going to market	Business model and go-to-market planning	商業模式與市場推出規劃

Exam takeaway

A strong answer should show that a good idea is not automatically an innovation. It needs refinement, evidence, a viable business model, and market acceptance.

高分答案要顯示：好點子不會自動成為創新；它需要優化、證據、可行商業模式與市場接受。

7. Creativity, Innovation, and the “Wow Effect” 創意、創新與「Wow Effect」

Key concepts

Concept	English	中文
Creativity	Producing something new, unusual, different, surprising, or unexpected.	產生新穎、不尋常、不同、令人驚喜或出乎意料的東西。
Innovation	Developing an idea into value that can be accepted by customers, society, or the market.	把想法發展成可被顧客、社會或市場接受的價值。
Wow effect / wow moment	The sense that an idea is unusual, exceptional, surprising, or unexpected.	一個想法帶來不尋常、出色、驚喜或意想不到的感覺。

- Both creativity and innovation begin with ideas. 創意與創新都從想法開始。
- However, not every idea is worth developing. 但並非每個想法都值得繼續發展。
- A potentially valuable idea often creates a “wow moment” because it is meaningfully different or surprising.
一個有潛力的想法，通常會帶來「wow moment」，因為它具有有意義的不同或驚喜感。
- The goal is to identify ideas worthy of further study, nurturing, support, and prototyping. 目標是找出值得進一步研究、培養、支持和製作原型的想法。

Creativity generates possibilities; innovation delivers accepted value.

創意產生可能性；創新交付被接受的價值。

Exam takeaway

When comparing creativity and innovation:

- Creativity focuses on novelty and possibility.
創意聚焦於新穎性與可能性。
- Innovation focuses on implementation and value realization.
創新聚焦於落實與價值實現。
- The “wow effect” helps identify ideas with potential, but the idea still needs development and market acceptance.
「Wow effect」有助識別具潛力的想法，但該想法仍需要發展及市場接受。

8. Why Study Types of Innovation? 為何要學習不同創新類型？

The purpose is not merely to memorize categories from a textbook. Studying innovation types helps managers understand how technology has evolved, recognize broad patterns, and make more informed monitoring, governance, and prediction decisions.

學習創新類型的目的是不只是背誦教科書分類，而是幫助管理者理解技術如何演進、辨識大致規律，以及作出更有根據的監察、管治和預測決策。

Key term: Technology trajectory 技術軌跡

English	中文
A technology trajectory refers to the broad developmental path along which a technology evolves over time.	技術軌跡是指一項技術隨時間發展時所走的大致路徑。
Past trajectories do not guarantee the future, but they help managers identify patterns and potential directions.	過去軌跡不能保證未來，但能幫助管理者辨識規律和潛在方向。

Exam takeaway

Use cautious language:

Past innovation patterns are useful guides, but they are not exact forecasts.

過去創新規律是有用的指引，但不是精確預測。

This is especially important because technology evolves quickly through more data, newer technologies, and lower costs.

這一點尤其重要，因為科技會因更多數據、更新技術與更低成本而迅速演進。

9. Incremental vs Radical/Disruptive Innovation 漸進式創新 vs 激進／顛覆式創新

Basic comparison

Dimension	Incremental innovation	Radical / disruptive innovation
English	Small, gradual, lower-risk improvements to what already exists.	Material and fundamental change from what existed before.
中文	對既有產品、服務或流程作小幅、逐步、較低風險的改善。	與既有做法相比出現重大及根本性的改變。
Typical approach	Steady, safe, step-by-step consolidation.	Greater uncertainty, larger departure from established practice.
常見做法	穩定、安全、逐步鞏固。	不確定性較高，與既有做法的差異較大。

- Most established firms are comfortable with incremental projects because they fit existing capabilities and routines.
大部分既有企業較習慣做漸進式專案，因為這些專案符合現有能力及工作慣例。
- However, relying only on incremental innovation creates vulnerability to new entrants and disruptive changes.
但若只依賴漸進式創新，企業會更容易受新進入者及顛覆性改變威脅。
- Firms need some projects in the radical space, even though they involve risk.
企業需要保留部分激進式創新項目，即使這些項目帶有風險。

Incremental protects today; radical prepares for tomorrow.

漸進式創新守住今天；激進式創新準備明天。

Exam takeaway

Do not argue that radical innovation is always better. A good answer explains the need for a portfolio balance:

- Exploit current strengths and profitability.
利用現有優勢與盈利能力。
- Explore emerging opportunities and possible disruptions.
探索新興機會與潛在顛覆。

10. Architectural Framework for Innovation Types 創新類型的架構框架

Framework logic

The lecture introduces a simplified framework based on two kinds of impact:

1. **Impact on components** — whether the individual parts change substantially.
對組件的影響 — 個別組成部分是否出現重大改變。
2. **Impact on links between components** — whether the way components connect is fundamentally rewired.
對組件之間連結的影響 — 組件之間的連接方式是否被根本重組。

Innovation types mentioned / 提及的創新類型

Type	English explanation	中文說明
Incremental	Small changes to components and small changes to their links.	組件及其連結都只有小幅改變。
Modular	More substantial change to components while the broader architecture is less radically changed; modularity can improve flexibility and versatility.	組件有較大改變，但整體架構未必完全重組；模組化可提升靈活性與多用途性。
Architectural	Fundamental reconfiguration of how components are connected.	根本性地改變組件之間的連接方式。
Radical	Significant change in both components and/or their underlying configuration, creating a major departure from the past.	組件及／或其基本配置出現重大改變，與過去做法有明顯斷裂。

Exam takeaway

When classifying a case, ask:

1. Did the **components** change significantly? 組件有否顯著改變？
2. Did the **relationships/links** among components change significantly?
組件之間的關係／連結有否顯著改變？

This lets you justify the innovation category logically rather than merely naming it.

這可讓你用邏輯去論證創新類型，而不是只寫出類別名稱。

11. Disruptive Innovation and the Entrant's Opportunity 顛覆性創新與新進入者的機會

Definition

Disruptive innovation involves entering a market where established firms are less willing or less able to respond, often because the new segment offers lower margins, does not fit their customers, or conflicts with their existing cost structures and processes.

顛覆性創新通常指新進入者進入一個既有企業不太願意或沒有能力回應的市場；原因可能是該市場利潤較低、不符合其目標客群，或與其成本結構及流程不相容。

Example 1: Toyota and premium car makers Toyota 與高級汽車品牌

English	中文
Premium car makers such as Mercedes and BMW were designed around high-priced vehicles, high cost structures, and high-margin markets.	Mercedes 和 BMW 等高級汽車品牌，是圍繞高價車型、高成本結構及高利潤市場而建立。
Toyota served a large segment that could not afford premium cars by offering more economical vehicles.	Toyota 透過提供較經濟型車款，服務大量買不起高級車的顧客。
Premium firms found it difficult to compete profitably in the lower-priced segment because their cost structures did not fit it.	高級品牌難以在低價市場中獲利競爭，因為其成本結構不適合該市場。
Toyota later moved upward with Lexus to compete closer to the premium segment.	Toyota 後來透過 Lexus 向上延伸，進一步接近高級車市場。

Example 2: Budget airlines 廉價航空公司

Budget airlines compete by unbundling the price:

- A basic ticket price is offered.
- Preferred seats, extra luggage, and other services cost extra.
- This attracts customers who mainly want low-cost transportation.

A smaller entrant should compete where the incumbent is least likely to respond. This usually happens when the incumbent sees the new market as unattractive, unfamiliar, low-margin, or inconsistent with its existing structure.

較小的新進入者應在既有企業最不可能回應的地方競爭。這通常發生在既有企業認為該市場不吸引、不熟悉、低利潤，或與其既有結構不相容的時候。

Exam takeaway

For disruptive innovation cases, explain both sides:

- **Entrant advantage:** targets overlooked customers or a segment incumbents avoid.
新進入者優勢：鎖定被忽略的顧客或既有企業迴避的市場。
- **Incumbent constraint:** existing processes, costs, customers, and profit logic make response difficult.
既有企業限制：既有流程、成本、顧客與盈利邏輯令回應困難。

12. Dynamic Capabilities and the Innovator's Dilemma 動態能力與創新者的兩難

Definition: Dynamic capabilities 動態能力

English	中文
Dynamic capabilities are the ability to rapidly reconfigure internal and external resources in response to environmental change.	動態能力是指企業因應環境變化，快速重組內部與外部資源的能力。

Definition: Innovator's dilemma 創新者的兩難

English	中文
The innovator's dilemma occurs when successful firms focus heavily on optimizing existing profitable products and overlook emerging niche disruptions.	創新者的兩難是指成功企業過度專注於優化既有高利潤產品，因而忽略新興利基市場的顛覆。
The problem is not necessarily incompetence; it is the rational tendency to protect current profits and customers.	問題未必是企業無能，而是企業理性地傾向保護當前利潤與既有顧客。

Managers face a permanent balance:

Exploitation	Exploration
Improve and profit from current assets, products, and capabilities. 利用並改善目前資產、產品與能力。	Search for risky, uncertain innovations needed for the future. 探索未來所需、帶有風險與不確定性的創新。

Too much exploitation creates the risk of being overtaken by disruption.

過度利用既有能力，可能會被顛覆者超越；

Too much exploration can exhaust capital before viable results are achieved.

過度探索，則可能尚未找到可行結果便耗盡資金。

Exploit today, explore tomorrow. 利用今天，探索明天

Exam takeaway

For an innovator's dilemma answer, include:

1. Existing success creates incentives to protect current profit streams.
現有成功會令企業傾向保護當前利潤來源。
2. New technologies may initially appear unattractive, small, or low-margin.
新技術在初期可能看似不吸引、規模小或利潤低。
3. The firm needs dynamic capabilities and a balanced innovation portfolio.
企業需要動態能力及平衡的創新組合。

13. Evolutionary Trap: Why Past Success Can Become Dangerous

Definition

An **evolutionary trap** happens when a firm repeats decisions that worked in the past, even though the external environment has changed and those decisions are no longer appropriate.

- Past cases and theories are useful, but they should not be copied mechanically.
過去案例和理論有用，但不能機械式照搬。
- A manager must reconsider the situation, revisit assumptions, redesign decisions, and sometimes depart from the earlier path.
管理者必須重新思考情境、檢視假設、重新設計決策，並在某些時候偏離舊路徑。
- History is a guide, not a window into the future.
歷史是指引，不是通往未來的窗口。

Example

A firm may have succeeded by using a certain strategy in an earlier technological environment. If competitors, customer expectations, cost conditions, or available technologies change, applying the same strategy again can lead to failure.

企業可能曾在舊科技環境中以某策略取得成功；但若競爭者、顧客期望、成本條件或可用技術已改變，重複同一策略便可能導致失敗。

Exam takeaway

Do not copy the past; diagnose the current context. 不要複製過去；要診斷當前情境。

Use this concept when a case shows overconfidence, rigid routines, or a failure to adapt to changed external conditions.

當案例顯示企業過度自信、僵化地遵循慣例，或未能適應已改變的外部環境時，可使用此概念。

Summary

- 1. Innovation is not just a new idea.** It is the process of converting ideas into accepted value and sustaining that value in the market.
創新不只是新點子。它是把想法轉化為被接受的價值，並在市場中維持該價值的過程。
- 2. Use systems thinking.** Firms operate through interconnected inputs, processes, outputs, and feedback loops; one change can affect the whole system.
使用系統思維。企業透過互相連結的輸入、過程、輸出和回饋循環運作；一個改變可影響整個系統。
- 3. Global forces accelerate innovation.** Fast technology adoption, Industry 4.0, cyber-physical systems, IoT, AI, and data abundance reshape how firms compete.
全球力量加速創新。快速科技採用、第四次工業革命、資訊物理系統、物聯網、AI 與數據豐裕，正在重塑企業競爭方式。
- 4. Data requires intelligence.** The managerial challenge is turning information abundance into useful insights and decisions through analytics and AI.
數據需要轉化為洞見。管理挑戰是透過分析與 AI，把資訊過量轉化為有用洞見與更佳決策。
- 5. Customer experience is stronger than satisfaction alone.** Satisfaction is transactional; experience creates loyalty through the full customer journey.
顧客體驗比單純滿意度更有力。滿意度是交易式的；體驗透過整個顧客旅程建立忠誠度。
- 6. Innovation moves from idea to market.** Ideas need insight, solution development, iteration, a business model, and market launch.
創新由想法走向市場。想法需要洞見、解決方案發展、迭代、商業模式及市場推出。
- 7. Creativity produces novelty; innovation produces value.** The “wow effect” helps identify surprising ideas that may deserve further development.
創意產生新穎性；創新創造價值。「Wow effect」有助識別值得進一步發展的驚喜想法。
- 8. Balance incremental and radical innovation.** Incremental projects protect current performance, but radical exploration prepares firms for disruption.
平衡漸進與激進創新。漸進式項目守住目前表現；激進探索則讓企業為顛覆作準備。
- 9. Disruptors often enter overlooked markets.** Incumbents may fail to respond because the segment does not fit their existing cost structures, processes, customers, or profit logic.
顛覆者常從被忽略的市場進入。既有企業未必能回應，因為該市場可能不符合其成本結構、流程、顧客或盈利邏輯。
- 10. Avoid the innovator’s dilemma and evolutionary trap.** Do not let current success or past decisions prevent adaptation; build dynamic capabilities and reconsider assumptions when the environment changes.
避免創新者的兩難與演化陷阱。不要讓當前成功或過去決策阻礙適應；當環境改變時，要建立動態能力並重新檢視假設。